[INCH-POUND]
A-A-59440/1A
23 May 2005
SUPERSEDING
A-A-59440/1
13 August 1999

COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

COCKS, DRAIN, CENTER DRAIN, INTERNAL SEAT OR EXTERNAL SEAT, 150 P.S.I., BRASS, NON-FLAMMABLE LIQUID OR GAS, NPTF-1 THREADS

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

The complete requirements for procuring the cocks, center drain, internal seat or external seat, 150 psi, brass, non-flammable liquid or gas, NPTF-1 threads described herein shall consist of this document and the issue in effect of CID A-A-59440.

CLASSIFICATION/PART OR IDENTIFICATION NUMBER (PIN). This CID uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see NOTES).

AA59440	/1	-	001	Blank or C
CID number	Applicable CID sheet (two digits)		Dash number (see figures 1 and 2)	Handle plating (see figures 1 and 2)

SALIENT CHARACTERISTICS.

Interface and physical dimensions. Cocks, center drain, internal seat or external seat, 150 psi, brass, non-flammable liquid or gas, NPTF-1 threads shall be as specified herein (see figures 1 and 2).

Sizes. Nominal sizes of 1/8 inch through 1/4 inch with NPTF-1 threads are covered by this CID.

Intended applications. These items are for use in non-flammable liquid or gas systems where pressures do not exceed 150 psig (10.3 bar) for water, oil and gas, (WOG) at 150°F (65.6°C) and the liquid is compatible with brass.

Permissible variations in dimensions. Figures 1 and 2 lists the permissible variations in dimensions for each drain cock size. The external dimensions of the drain cocks should not limit procurement unless the end item applications restrict these dimensions.

Optional designs. Figures 1 and 2 are for identification purposes only and are not intended to restrict designs or shapes not dimensioned.

Valve bodies. Valve bodies shall be provided with hexagonal wrenching surfaces.

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Threads. Threads are Dryseal NPTF-1 in accordance with CID A-A-59440.

Material.

Brass. Body and handle, external seat only, shall be brass in accordance with the following commercial standards:

ASTM B16	ASTM B121	ASTM B505	SAE AMS 4842
ASTM B22	ASTM B124	ASTM B584	SAE AMS 4845
ASTM B30	ASTM B148	ASTM B763	SAE AMS 4855
ASTM B36	ASTM B176	ASTM B770	SAE AMS 4860
ASTM B61	ASTM B271		SAE AMS 4862
ASTM B62	ASTM B369		SAE AMS 4890

Handle internal seat (see figure 1). Handle may be brass, rolled or cold rolled steel conforming to QQ-S-698 or another commonly accepted industry standard.

Handle external seat (see figure 2). Handle may be hot rolled or cold rolled steel conforming to QQ-S-698 or another commonly accepted industry standard.

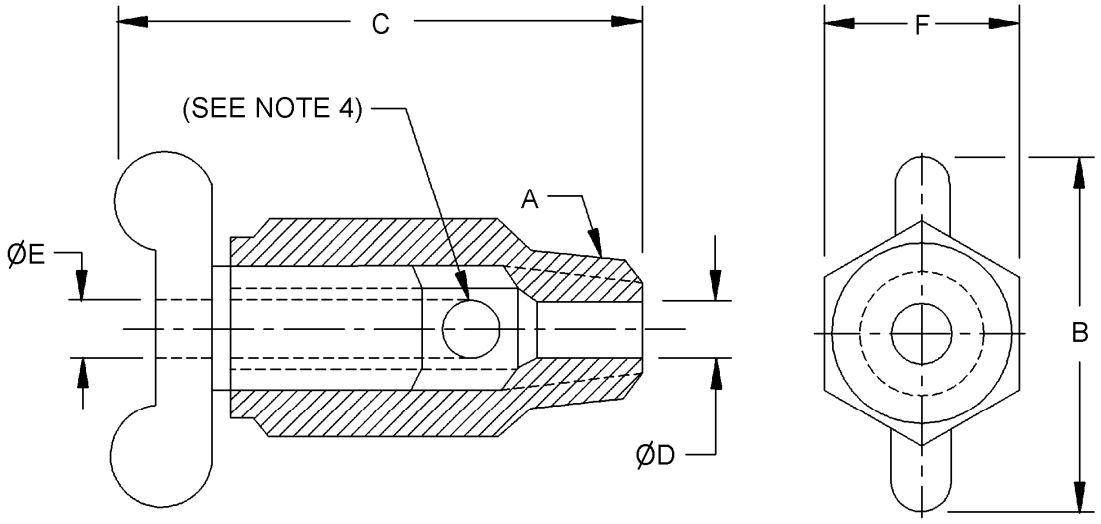
Protective coating:

a. Brass: None.
b. Carbon steel:
   (1) Cadmium in accordance with SAE AMS-QQ-P-416, type II, class 2.
   (2) Zinc in accordance with ASTM B633, type II, 12.7 µm (50 microinches) minimum.

Test pressures:

a. Shell: 150 psig (10.3 bar)
b. Seat: 50 psig (3.45 bar).

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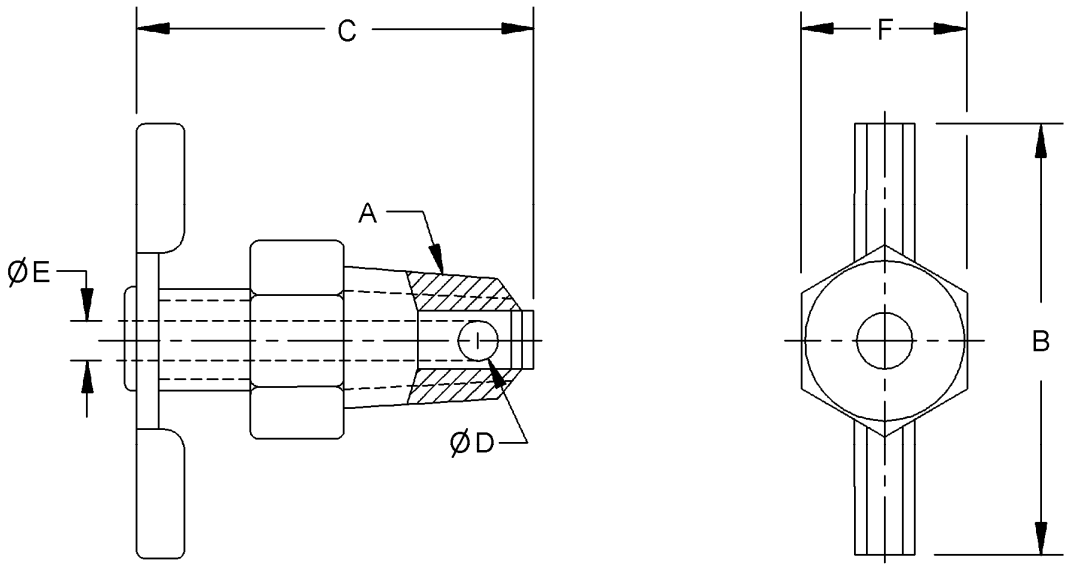
INTERNAL SEAT

Dash No.	Handle plating	A Pipe Thread NPTF-1	B max (mm)	C open max. (mm)	D dia. min. (mm)	E dia. min. (mm)	F ±.015 (0.38 mm)
001	Zinc or Brass	1/8-27	1.38 (34.0)	1.88 (47.8)	.19 (4.8)	.12 (3.1)	.44 or .56 (11.2 or 14.2)
001C	Cadmium						
002	Zinc or Brass	1/4-18	1.50 (38.1)	2.00 (50.8)	.19 (4.8)	.19 (4.8)	.56 or .62 (14.2 or 15.7)
002C	Cadmium						
003	Zinc or Brass	3/8-18	1.50 (38.1)	2.00 (50.8)	.31 ±.02 (7.6 ± 0.5)	.25 to .31 (6.4 to7.9)	.69 or .75 (17.5 or 19.1)
003C	Cadmium						

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ± .010 inch (0.25 mm).
4. Diameter of hole shall be adequate to allow drainage through the valve.

FIGURE 1. Dimensions and configuration internal seat.

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EXTERNAL SEAT

Dash No.	Handle plating	A pipe thread NPTF-1	B max (mm)	C closed ref (mm) 4/	D dia ref (mm) 5/	E dia min (mm)	F ±.015 (0.38 mm)
004	Zinc	1/8-27	1.31 (33.3)	.90 (22.86)	.09 (2.3)	.12 (3.0)	.44 or .50 (11.2 or 12.7)
004C	Cadmium						
005	Zinc	1/4-18	1.38 (35.1)	1.00 (25.4)	.16 (4.1)	.19 (4.8)	.50 or .56 (12.7 or 14.2)
005C	Cadmium						
006	Zinc	3/8-18	1.69 (42.93)	1.44 (36.6)	.16 (4.1)	.19 (4.8)	.62 or .69 (15.7 or 17.5)
006C	Cadmium						
007	Zinc	1/2-14	2.00 (50.8)	1.62 (41.1)	.31 (7.9)	.38 (9.7)	.81 or .88 (20.6 or 22.4)
007C	Cadmium						
008	Zinc	1-11 1/2	2.50 (63.5)	2.31 (58.7)	.53 (13.5)	.75 (19.1)	1.38 or 1.50 (35.1 or 38.1)
008C	Cadmium						

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ±.010 inch (0.25 mm).
4. Dimension “C” open/closed length dimensions may vary from manufacturer to manufacturer.
5. Diameter of hole shall be adequate to allow drainage through the valve.

FIGURE 2. Dimensions and configuration external seat.

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NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID. See classification information for PIN format example.

Source of documents.

FEDERAL SPECIFICATION

QQ-S-698	-	Steel, Sheet and Strip, Low Carbon

COMMERCIAL ITEM DESCRIPTION

A-A-59440	-	Cocks, Drain, Cocks, Plug, and Cocks, Shutoff, Screw Stem; Brass, NPT and NPTF-1 Threads, General Requirements For

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

Other Publications

ASTM INTERNATIONAL

ASTM B16	-	Rod, Brass, Free-Cutting, Bar and Shapes for use in Screw Machines
ASTM B22	-	Castings, Bronze, for Bridges and Turntables
ASTM B30	-	Copper Alloys in Ingot Form
ASTM B36	-	Plate, Brass, Sheet, Strip, and Rolled Bar
ASTM B61	-	Castings, Steam or Valve Bronze
ASTM B62	-	Metal Castings, Composition Bronze or Ounce
ASTM B121	-	Plate, Leaded Brass, Sheet, Strip, and Rolled Bar
ASTM B124	-	Copper and Copper Alloy Forging Rod, Bar, and Shapes
ASTM B148	-	Aluminum-Bronze Sand Castings
ASTM B176	-	Die Castings, Copper-Alloy
ASTM B271	-	Copper-Base Alloy Centrifugal Castings
ASTM B369	-	Copper-Nickel Alloy Castings
ASTM B505	-	Castings, Copper Alloy, Continuous
ASTM B584	-	Castings, Copper Alloy Sand, for General Applications
ASTM B633	-	Zinc on Iron and Steel, Electrodeposited Coatings of
ASTM B763	-	Copper Alloy Sand Castings for Valve Application
ASTM B770	-	Copper-Beryllium Alloy Sand Castings for General Applications

(Copies of these documents are available from http://www.astm.org or ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, Pennsylvania, USA 19428-2959.)

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SAE INTERNATIONAL

SAE AMS4842	-	Castings, Leaded Bronze, Sand and Centrifugal 80CU - 10SN - 9.5PB as Cast
SAE AMS4845	-	Tin Bronze Castings, Sand and Centrifugal 87.5CU - 10SN - 2ZN as Cast
SAE AMS4855	-	Leaded Red Brass, Sand and Centrifugal Castings 85CU - 5.0SN - 5.0PB - 5.0ZN as Cast
SAE AMS4860	-	Manganese Bronze, Sand and Centrifugal Castings 58CU - 39ZN - 1.2FE - 1.0AL - 0.80MN as Cast
SAE AMS4862	-	Manganese Bronze, Sand and Centrifugal Castings 63CU - 24ZN - 6.2AL - 3.8MN - 3.0FE High Strength, as Cast
SAE AMS4890	-	Copper-Beryllium Alloy Castings 97CU - 2.1BE - 0.52CO - 0.28SI Solution Heat Treated (TBOO)
SAE-AMS-QQ-P-416	-	Plating, Cadmium (Electrodeposited)

(Copies of these documents are available from http://www.sae.org/ or SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

Commercial products. As part of the market analysis and research effort, this CID was coordinated with the following manufacturers of commercial products. At the time of CID preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

MFR's CAGE	MFR's name and address
01276	Eaton Aeroquip Inc. 14615 Lone Oak Road. Eden Prairie, MN 55344 952-294-2032/888-258-0222 jeffenoe@eaton.com
93061	Parker-Hannifin Corp Brass Products Division 300 Parker Drive Otsego, MI 49708-1472 260-636-2104 dwsanders@Parker.com
20969	Essex Brass Corp 23500 Pinewood Warren, MI 48091-3122 586-757-8200 bpushman@essexbrass.com

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Military Standard (MS) replacement data. This CID replaces the following Military Standard as shown in table I.

TABLE I. MS supersession data.

CID PIN 1/	Superseded MS
AA559440/1-001	MS35782-1
AA559440/1-001C	MS35782-1
AA559440/1-002	MS35782-2
AA559440/1-002C	MS35782-2
AA559440/1-003	MS35782-3
AA559440/1-003C	MS35782-3
AA559440/1-004	MS35782-4
AA559440/1-004C	MS35782-4
AA559440/1-005	MS35782-5
AA559440/1-005C	MS35782-5
AA559440/1-006	MS35782-6
AA559440/1-006C	MS35782-6
AA559440/1-007	MS35782-7
AA559440/1-007C	MS35782-7
AA559440/1-008	MS35782-8
AA559440/1-008C	MS35782-8

1/ These parts were originally listed in MS35782 as a single part type. Due to special handling/disposal and corrosion resistance requirements the parts have been separated in this CID.

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Part number (P/N) supersession data. These CID P/N's supersede the following MFR's P/Ns as shown in table II. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE II. P/Ns Supersession data. 1/

CID PIN	MFR's CAGE 2/	MFG's P/N	MFR's CAGE	MFG's P/N	MFR's CAGE	MFG's P/N
AA59440/1-001	01276	130	20969	DC100 1/8	93061	DC603-2
AA59440/1-001C						
AA59440/1-002	01276	140	20969	DC100 1/4	93061	DC603-4
AA59440/1-002C						
AA59440/1-003	01276	230		N/A	93061	DC603-6
AA59440/1-003C				N/A		
AA59440/1-004	01276	135		N/A	93061	DC604-2
AA59440/1-004C				N/A		
AA59440/1-005	01276	145		N/A	93061	DC604-4
AA59440/1-005C				N/A		
AA59440/1-006	01276	270		N/A	93061	DC604-6
AA59440/1-006C				N/A		
AA59440/1-007	01276	108		N/A	93061	N/A
AA59440/1-007C				N/A		N/A
AA59440/1-008	01276	N/A		N/A	93061	N/A
AA59440/1-008C		N/A		N/A		N/A

1/ The manufacturer's PIN shall not be used for procurement to the requirements of this CID. At the time of preparation of this CID, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements see marking paragraph.

2/ Manufacturers products not approved for steam service

National stock number (NSN). The following is a list of NSN's assigned which correspond to this CID. The list in table III is for information only and may not be indicative of all possible NSN's associated with the CID. For up to date information on assigned NSN's please contact the aforementioned DSCC office (see A-A-59440).

TABLE III. NSN's

Dash number (see figure 1) AA59440/1-	NSN
-003	4820-00-174-0339
-004	4820-00-752-9040
-005	4820-00-849-1220
-007	482001-229-1724

Government users. To acquire information on obtaining these cocks from the Government inventory system, contact Defense Supply Center, Columbus, ATTN: DSCC-NAB, P.O. Box 3990, Columbus, OH 43218-3990, or telephone (614) 692-2271 or (614) 692-3191.

Marginal notations. Marginal notations are not used in this revision to identify changes due to the extent of the changes.

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MILITARY INTERESTS:

Custodians:
Army - AR
Navy - SH
Air Force - 99
DLA - CC

Review activities:
Army - AT, AV, CR4, GL, MI
Navy - MC

CIVIL AGENCY AND COORDINATING ACTIVITIES:

GSA - FSS

Preparing activity

DLA - CC

Project 4820-0881-000

NOTE: The activities listed above were interested in this document as of the date of this document. Since organization and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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